

Waste Image Classification using PyTorch

1. Problem Statement

Waste segregation is an important problem in waste management. Different types of waste need to be classified properly for efficient recycling and disposal.

Manual waste classification can be time-consuming and inconsistent.

The goal of this project is to build an **image classification system using PyTorch** that can identify the category of waste from an input image.

Given an image of a waste item, the system will use a deep learning model to predict its waste category.

2. Objective

The objective of this project is to apply the PyTorch concepts learned so far to a practical computer vision problem.

The project will focus on understanding how a PyTorch-based image classification model works, including:

- Preparing image data
 - Loading data using PyTorch
 - Building a neural network
 - Training the model
 - Calculating loss
 - Using backpropagation
 - Updating model parameters
 - Evaluating the model
 - Making predictions on images
-

3. Type of Problem

Problem Type: Image Classification

Domain: Computer Vision

Framework: PyTorch

Programming Language: Python

The model will classify an input image into one of the waste categories defined by the dataset.

4. PyTorch Concepts Already Learned

The following concepts have already been studied and practiced:

PyTorch Fundamentals

- Tensors
- Tensor creation
- Tensor operations
- Tensor shapes
- Matrix multiplication
- Automatic differentiation
- `requires_grad`
- Gradients
- `backward()`

Model Training

- Forward pass
- Loss functions
- Backpropagation
- Optimizers
- `zero_grad()`
- `optimizer.step()`
- Training loops

Neural Networks

- Neural network layers
- `nn.Module`
- `nn.Linear`
- Activation functions
- ReLU
- `nn.Sequential`

Dataset and Training Workflow

- Training data
- Testing data
- Dataset
- DataLoader
- Batches
- Epochs
- Training mode
- Evaluation mode

Model Evaluation

- Loss
- Accuracy
- Predictions
- `torch.no_grad()`

- `model.eval()`

Model Saving

- Saving PyTorch models
- Loading saved model parameters
- Using a trained model for inference

Computer Vision

- Image classification
 - CNN concepts
 - Convolution
 - Feature extraction
 - Pooling
 - Max pooling
 - Average pooling
-

5. Previous PyTorch Practice

Before starting this project, PyTorch was practiced through smaller problems and projects.

Polynomial Regression

A polynomial regression model was implemented using PyTorch.

The workflow included:

```
Input Data
  ↓
Prediction
  ↓
MSE Loss
  ↓
Backpropagation
  ↓
Optimizer
  ↓
Weight Update
```

MNIST Image Classification

A neural network was implemented for handwritten digit classification.

The model used:

```
784 input features
  ↓
```

```
Linear Layer
  ↓
ReLU
  ↓
Linear Layer
  ↓
ReLU
  ↓
Linear Layer
  ↓
10 output classes
```

The model was trained using:

- Cross Entropy Loss
- Adam optimizer
- Multiple epochs
- Training and evaluation loops

The trained model was also used for individual image prediction.

6. Current Project

Project Name: Waste Image Classification using PyTorch

Current Status: Project planning / problem definition

The PyTorch fundamentals required to begin this project have already been learned and practiced.

The implementation will be developed based on the concepts already studied rather than following a tutorial step-by-step.

7. Learning Goal

The main purpose of this project is to move from small PyTorch practice projects to a practical computer vision project.

The core learning workflow is:

```
Image Data
  ↓
PyTorch
  ↓
Neural Network
  ↓
Training
  ↓
```

Loss



Backpropagation



Parameter Updates



Evaluation



Prediction